A Structural Analysis of Parameter Reduction in Left-Branch Dynamics of Assani-Type Collatz Trees

Author: Chen Mao | **Email:** savy.ch@gmail.com

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Abstract

This note studies a parameter-reduction phenomenon arising in the analysis of left-branch trajectories in Assani-type Collatz trees. For the parameterized family $L_i=3^{k-i}2^i h-1$, we prove the deterministic relation between left-branch nodes and the root parameter root=$3^k h-1$. After reaching the root, we introduce an auxiliary escape transformation obtained from the odd transition followed by maximal removal of powers of two. Under a specific valuation condition, this transformation produces a reduced representation with a single visible factor of 3. We show that this reduction is a genuine algebraic phenomenon, but inherently fails to satisfy dynamical closure due to an algebraic obstruction under the base coprimality constraints. We provide an initial congruence classification of the escape valuation and formulate the invariant closure property required for a complete descent theorem.

1. Introduction

The Syracuse form of the Collatz map is defined by:

$$T(x) = x/2 \quad \text{if } x \equiv 0 \pmod 2$$
$$T(x) = (3x+1)/2 \quad \text{if } x \equiv 1 \pmod 2$$

We consider the parameter family **root = 3^kh-1**, where $k \geq 1$, $h \geq 1$, with h odd and 3 ■ h. Since $3^k h$ is odd, root is always even.

The associated left-branch nodes are defined as:

$$L_i = 3^{k-i}2^i h - 1, \qquad 0 \leq i \leq k$$

The purpose of this note is to analyze the structural parameter transformation occurring after a trajectory reaches the root.

2. Deterministic Left-Branch Dynamics

Proposition 2.1. For every $1 \leq i \leq k$, we have $T(L_i) = L_{i-1}$.

Proof. Because L_i is odd, applying the odd component yields:

$T(L_i) = (3L_i + 1) / 2 = [3(3^{k-i}2^i h - 1) + 1] / 2$

$= (3^{k-i+1}2^i h - 2) / 2 = 3^{k-i+1}2^{i-1}h - 1 = L_{i-1}$. ∅

Corollary 2.2. Repeated iteration gives $T^i(L_i)$ = root. Therefore, every left-branch node deterministically reaches its associated root.

3. Auxiliary Escape Transformation

Define the auxiliary escape operator:

$$F(N) = (3N - 2) / 2^{v_2(3N-2)}$$

This operator acts directly on the even root parameter as a composite algebraic extractor, bypassing the initial division by 2 to analyze the subsequent odd-step parity boundary. For N = root = $3^k h - 1$, we get $3N - 2 = 3^{k+1}h - 5$. Hence, the parameter identity is exact:

$$F(N) = (3^{k+1}h - 5) / 2^{v_2(3^{k+1}h - 5)}$$

4. Visible 3-Factor Reduction

Assume the primary non-vanishing residue condition $v_2(3^{k+1}h-5) = 1$. Then:

$$F(N) = (3^{k+1}h - 5) / 2$$

Adding one to both sides yields $F(N) + 1 = (3^{k+1}h - 3) / 2 = 3 \times [(3^k h - 1) / 2]$. Let $h' = (3^k h - 1) / 2$. Then we establish the mapping:

$$F(N) = 3h' - 1$$

Thus, the transformed state has a representation with exactly one explicit factor of 3: $F(N) = 3^1 h' - 1$. This establishes the parameter-reduction phenomenon.

4.1 Initial Mod-8 Classification of the Escape Valuation

k parity	h (mod 8)	Initial Valuation Info
k even	$h \equiv 1, 5$	v = 1 (Reduction Triggered)
k even	$h \equiv 3$	v = 2
k even	$h \equiv 7$	$v \geq 3$ possible
k odd	$h \equiv 3, 7$	v = 1 (Reduction Triggered)
k odd	$h \equiv 1$	v = 2
k odd	$h \equiv 5$	$v \geq 3$ possible

5. Limitation of Parameter Reduction

The representation $F(N) = 3h' - 1$ has the same algebraic form as a k=1 root, but it fails to generate dynamical closure into the base class. The standard k=1 left-branch family requires nodes to be of the form $L_1 = 2h - 1$. Therefore, $3h' - 1$ does not generally belong to the previously analyzed left-branch class. Hence, **parameter reduction does not imply dynamical closure**.

6. General Escape Structure

For arbitrary v, $F(N) = (3^{k+1}h - 5) / 2^v$. The information is contained in $F(N) + 1 = (3^{k+1}h - 5 + 2^v) / 2^v$. Since $v_3(2^v) = 0$, the subsequent 3-adic exponent depends strictly on the congruence structure of $3^{k+1}h - 5 + 2^v \pmod{3^r}$, which exhibits chaotic dispersion.

7. Conditional Descent Framework

Definition 7.1 (Escape Closure). Let C be a class of parameterized states. We say that C satisfies escape closure if $N \in C \Rightarrow F(N) \in C_{desc}$, where every element of C_{desc} has an established descent property.

Theorem 7.2 (Conditional Left-Branch Descent). Assume every left-branch node reaches its root, and the escape transformation maps roots into a known descending class. Then every left-branch node eventually descends below itself.

8. Examples of Non-Closure and Structural Obstruction

The visible reduction does not imply membership in the left-branch family due to a strict coprimality contradiction. For $3h' - 1$ to belong to the k=1 left branch, we need $3h' - 1 = 2h\sim - 1 \Rightarrow h\sim = 3h' / 2$.
1. If h' is odd, $h\sim \notin Z$ (contradiction).
2. If h' is even, h~ is an integer, but $3 \mid h\sim$, which violates the foundational constraint 3 ■ h.

9. Open Problem

Does there exist a natural invariant class C such that $F(C) \subseteq C_{desc}$? A positive answer would convert the observed parameter reduction into a complete descent mechanism.

10. Summary

Statement	Status
Left-branch identity $T(L_i) = L_{i-1}$	Proven
Root reaching property	Proven
Escape transformation formula	Proven
Visible 3-factor reduction	Proven
Reduction to k=1 left branch	False in general
Complete left-branch descent	Conditional